

## KEY DEVELOPMENT

# SIEGES AND THE ART OF FORTIFICATION

**Castles and siege warfare flourished in the medieval period, mainly as a result of widespread insecurity and the fragmentation of political power. As the technology of siege weapons and tactics evolved, so it drove developments in the design of castles and city walls.**

In the 11th century, most castles in western Europe were still made of wood and earth. When the first crusader armies from the west made their way to the Holy Land in 1096–99, they were awed by the defensive structures they found – massive stone fortifications ringed the great cities around the eastern Mediterranean, such as Constantinople and Jerusalem. During the following century, stone castles were to become the norm across western Europe, as well as in the crusader states in the east. At first these were mostly quite simple, with a tall curtain wall surrounding an elevated stone keep. This posed sufficient problems for besieging forces equipped with mangonels – stone-throwing torsion catapults inherited from the Romans. Unable to create a breach in the walls, armies mostly settled down to a long blockade.

## INCREASED DEFENCES

The introduction of the counterweight trebuchet in the 12th century changed the balance between besiegers and besieged. These huge machines were capable of hurling rocks weighing over 100kg (220lbs), and made curtain walls vulnerable. Castle designers responded by creating concentric structures, with the outer wall merely as a first line of defence, behind which the even taller, thicker fortifications of an inner castle loomed. Towers in the battlements were made rounded, instead of square, to deflect the force of a hurled rock. A castle was surrounded by moats – either dry or filled with water – and fortified outworks to make it difficult for the besiegers to bring up rams or siege towers. Potential weak points, especially the main gate, were reinforced with extra fortifications. Arrow slits in the walls and towers were positioned to shoot directly onto troops advancing towards, or successfully breaching, the castle.

## SIEGE LOGISTICS

A medieval siege was a major undertaking, and even maintaining a blockade could stretch an army's resources. Spread out around the fortified position, besiegers were vulnerable to sudden sorties by the defenders' garrison, as well as to sniping from crossbowmen on the battlements. If

mangonels and trebuchets failed to make a breach in the walls, these could be attacked with rams, or by tunnelling to undermine the foundations. A portable roof – the cat – was deployed to protect troops against missiles from above as they advanced to bring forward a ram, or hack at the wall with picks. Occasionally an assault on unbreached walls succeeded, either by scaling ladders or by rolling a tall wooden siege tower forward. More often, a weak point such as an unlocked door was discovered – or the defenders were betrayed by treachery on the part of one of their own side.

The introduction of cannon became a transforming influence on sieges during the 15th century. French superiority in gunpowder artillery enabled them to overcome a series of English stone strongholds rapidly, in the 1440s and 1450s. The stone castle was soon rendered obsolete, but another revolution in the design of fortifications would soon make sieges as long and arduous as before – the era of the angle bastion (see pp.176–77).



## KEY BATTLE

## THE SIEGE OF ROUEN 1418–19

In July 1418, the city of Rouen, in France, was besieged by an English army under King Henry V. French crossbowmen held the attackers at bay, while primitive English cannon failed to batter a breach in the well-built masonry. After six months of tight blockade, the defenders were starved into submission.



▲ Painted in around 1480, this French image of the siege shows the attackers attempting to breach the city walls with their guns.

### ◀ MANGONEL

The catapult used the pulling power of a skein of twisted ropes. When released, the arm flew upwards and hurled the projectile at its target – often a castle's fortified walls.

### ▼ CRUSADER CASTLE

The castle of Krak des Chevaliers, of the long-standing religious and military order the Knights Hospitaller (1099–present), was considered one of the world's strongest fortresses. In 1271, it fell to Muslim leader Baybars, who used it as a base for attacks on crusader strongholds on the coast.

